NEURAL SPACETIME SEARCH WITH GRTECLYN: A PRACTICAL BASELINE FOR DYNAMICAL METRIC DISCOVERY

NIKITA M. SHIROKOV*
Independent Researcher, Moscow, Russia
(Dated: May 31, 2026)

We outline a practical baseline for turning **GRTeclyn** into a closed-loop GPU search engine for exotic spacetime metrics. A Python scheduler proposes compact initial-data parameters, a C++/AMReX executable evolves each candidate, and scoring diagnostics promote only geometries that survive, show operational faster-than-light (FTL) benefit, and avoid coordinate artifacts. Matter is handled either by algebraic reconstruction from the Einstein constraints at $t = 0$ or, in the long-term path, by an elliptic positive-energy initial-data solve.

I. INTRODUCTION

Most proposed interstellar metrics are hand-designed static ansätze. That approach is analytically clean but strongly biased toward highly symmetric geometries.

Here metric discovery is treated as black-box optimization over dynamical 3D initial data. The framework couples a Python optimizer to the GPU AMR evolution engine in `GRTeclyn`, with explicit checks for matter consistency, stability, and fake-FTL gauge effects.

This fits a broader shift toward automated theory exploration [1–4]. For warp geometries, WARP FACTORY and WARPAX emphasize observer-robust energy-condition evaluation [5, 6]; we extend that philosophy from analysis of prescribed metrics to search over evolved initial data, while keeping numerical convergence and constraint satisfaction central [7, 8].

Throughout this draft, "FTL" means effective faster-than-light propagation through a curved spacetime geometry compared with flat-spacetime light travel. It does not imply local superluminal motion relative to the local light cone.

II. GOAL

The goal is an automated numerical-relativity laboratory for FTL transport, FTL communication, portal-like shortcuts, and wormhole throats. A successful candidate must survive evolution, show an operational shortcut relative to a flat control, keep tidal forces bounded, and minimize violations of the standard energy conditions.

A. Optimization as the Engine of Discovery

Blind random 3D initial data almost always violates constraints, collapses, or relaxes to flat space. Optimization supplies the steering: maximize FTL benefit while penalizing constraint error, negative energy, tidal force, and instability.

* shirokov.nm@phystech.edu

III. INITIAL DATA AND MATTER DERIVATION

The initial spatial metric γ_{ij} and extrinsic curvature K_{ij} must satisfy the Einstein constraints, which determine the compatible energy density ρ and momentum density j_i. We use two matter-sector paths:

A. Path A: Algebraic Matter Reconstruction

The optimizer proposes geometry; the constraints give its required matter cost,

$$\rho = \frac{R + K^2 - K_{ij}K^{ij}}{16\pi}, \qquad j_i = \frac{1}{8\pi} D_j \left(K_i^j - \delta_i^j K \right), \tag{1}$$

with R the 3D Ricci scalar and D_j the covariant derivative. Regions with $\rho < 0$ are recorded as exotic-matter cost and fed back into the objective.

B. Path B: Elliptic Constraint Solving

The longer-term positive-energy path fixes a matter model with $\rho \geq 0$ and uses an elliptic solver (e.g. `GRTresna`) to deform each proposed geometry until the constraints are satisfied before evolution.

IV. CANDIDATE PARAMETERIZATION

The optimizer emits a compact vector θ; a decoder maps it to smooth AMReX initial fields.

A. Level 1: Radial Recipes

We define the initial fields at $t = 0$ using spherically symmetric radial profiles:

$$q(r) = q_\infty + \sum_n a_n B_n(r), \tag{2}$$

where q denotes χ, α, β^i, or K, and B_n are smooth radial basis functions.

B. Level 2: Angular Recipes (implemented)

Angular modes add non-spherical structure:

$$q(r,\theta,\varphi) = q_\infty + \sum_{n\ell m} a_{n\ell m} B_n(r) Y_{\ell m}(\theta,\varphi). \tag{3}$$

This is implemented for gauge fields (α, β^i), where angular modes carry no constraint cost; geometric angular modes require the future 3D constraint solver.

C. Levels 3–4: Neural and Meta-Optimization (future)

Future Level 3 replaces basis functions with an implicit neural decoder, optionally trained with Hamiltonian and momentum residuals as PINN losses [8].

a. A symbolic "spacetime grammar" (neuro-symbolic constraint filtering). A rule-based spacetime grammar can enforce asymptotic flatness, positivity, $K = \text{const}$, $\Pi = 0$, and chosen topology before evaluation, reducing wasted GPU jobs in the spirit of grammar-constrained theory generation [1].

b. Level 4: a concrete LLM meta-agent. Level 4 is an LLM meta-agent that classifies failed runs, retunes exposed gauge/grid parameters, and proposes new modes while leaving the physics verdict to read-only diagnostics [2].

V. GPU CLUSTER ARCHITECTURE AND PROGRAM BOUNDARY

The pipeline keeps Python orchestration separate from precompiled GPU evolution: Python proposes and scores candidates; the C++ executable only loads parameters, initializes fields, evolves, and writes diagnostics.

A. The Execution Boundary

The Python controller writes `params.txt` and dispatches Slurm tasks; GPU nodes run the same `GRTeclyn` binary for every candidate, avoiding recompilation inside the search loop.

B. Tiered GPU Execution

Tier 1 uses short low-resolution single-GPU runs with early stopping; Tier 2 promotes survivors to multi-GPU resolution ladders for stability and convergence.

VI. THE CLOSED-LOOP: HOW SIMULATION RESULTS DRIVE THE SEARCH

Each optimizer evaluation follows:

1. propose basis coefficients θ;
2. derive constrained fields, write `params.txt`, and reject bad pre-flight residuals;
3. evolve on GPU and write constraint, collapse, matter, and plotfile diagnostics;
4. parse diagnostics into $S(\theta)$ and update CMA-ES after each generation.

The optimizer sees only the scalar black-box objective $f(\theta) = -S(\theta)$, making the score design the central physics interface.

VII. SCORING: OVERCOMING THE FLAT-SPACE ATTRACTOR

The score must reward FTL-relevant structure without letting trivial Minkowski win by being perfectly stable and constraint-clean. The upgraded objective therefore adds continuous FTL scaffolding ($F_{\text{asymmetry}}$, $F_{\log}$), co-moving stability, growth penalties, and a non-triviality gate.

A. Implemented score components

Each component s_k is mapped through

$$r(v;\,\sigma) = \frac{1}{1 + v/\sigma}, \tag{4}$$

and combined as

$$S(\theta) = \sum_k w_k\, s_k(\theta). \tag{5}$$

B. Continuous FTL scaffolding metrics

Two continuous proxies give CMA-ES a gradient before a fully superluminal candidate appears.

1. *Natário–Alcubierre expansion asymmetry*

For a warp-like contraction-ahead/expansion-behind pattern, use

$$F_{\text{asymmetry}} = \max\left(0,\ \frac{\int_{x>0} -\theta(\mathbf{x})\, d^3x + \int_{x<0} \theta(\mathbf{x})\, d^3x}{\int |\theta(\mathbf{x})|\, d^3x + \epsilon_\theta}\right). \tag{6}$$

The implemented 1D evaluator reduces these integrals to trapezoidal sums along the travel axis.

2. Logarithmic FTL amplification

Weak shortcuts are log-amplified:

$$F_{\log} = \frac{\ln(1 + \eta F_{\mathrm{FTL}})}{\ln(1 + \eta)}, \tag{7}$$

with $\eta \approx 100$, so even a few-percent shortcut becomes optimizer-visible.

C. Co-moving stability scoring

Eulerian stability penalizes moving bubbles. We therefore estimate stability in a co-moving frame.

The average axial shift is

$$\langle \beta^x \rangle = \frac{1}{V_{\mathrm{bubble}}} \int_{\mathrm{bubble}} \beta^x(\mathbf{x})\, d^3x. \tag{8}$$

and the co-moving budget is

$$\begin{aligned} \Delta_{\mathrm{comoving}} &= \frac{\max_t |\partial \chi / \partial t|}{\max(|\langle \beta^x \rangle|,\, 10^{-3})}, \\ s_{\mathrm{comoving}} &= \frac{1}{1 + \Delta_{\mathrm{comoving}}}. \end{aligned} \tag{9}$$

For stationary throats or portals with negligible shift, `score.py` falls back to Eulerian stability.

Unavailable diagnostics contribute zero, so legacy episodes re-score consistently.

1. The non-triviality gate: defeating the flat-space attractor

Health rewards are multiplied by a non-triviality gate so flat space cannot win solely by being clean:

$$\begin{aligned} g_{\mathrm{nt}} = \min\Big(1,\ \max\big(&s_{\mathrm{nonflat}}, F_{\mathrm{asymmetry}}, s_{\mathrm{nontrivial}}, \\ &F_{\mathrm{curv}}, F_{\mathrm{op}}, F_{\log}, s^{\mathrm{eff}}_{\mathrm{exotic}}\big)\Big) \in [0, 1], \end{aligned} \tag{10}$$

The health rewards $\mathcal{H}$ are gated, while FTL/exoticity terms remain ungated:

$$S(\theta) = g_{\mathrm{nt}}(\theta) \sum_{k \in \mathcal{H}} w_k s_k(\theta) \ + \sum_{k \notin \mathcal{H}} w_k s_k(\theta). \tag{11}$$

With this gate, flat Minkowski drops to $S = 0$ while structured reference geometries keep their original scores.

D. FTL shortcut metrics (implemented)

`ftl_metrics.py` reconstructs $t = 0$ profiles and evaluates three shortcut indicators on the x-axis.

1. Warp null-ray shortcut

For conformally flat slices, an axial null ray has

$$\frac{dx}{dt} = \alpha(x)\sqrt{\chi(x)} - \beta^x(x). \tag{12}$$

with travel time

$$T_{\mathrm{curved}} = \int_{-L}^{L} \frac{dx}{\alpha\sqrt{\chi} - \beta^x}, \qquad T_{\mathrm{flat}} = 2L, \tag{13}$$

and reward

$$F_{\mathrm{null}} = \max\left(0,\ \frac{T_{\mathrm{flat}} - T_{\mathrm{curved}}}{T_{\mathrm{flat}}}\right). \tag{14}$$

Paths below a coordinate-speed floor are rejected.

2. Portal proper-distance shortcut

The portal-like compression reward is

$$\begin{aligned} D_{\mathrm{proper}} &= \int_{-L}^{L} \chi^{-1/2}(r)\, dr, \\ F_{\mathrm{portal}} &= \max\left(0,\ \frac{2L - D_{\mathrm{proper}}}{2L}\right), \end{aligned} \tag{15}$$

3. Wormhole throat pinch

The throat reward compares an interior areal-radius minimum to the asymptotic radius:

$$F_{\mathrm{throat}} = \max\left(0,\ 1 - \frac{R_{\min}}{R_{\mathrm{asymp}}}\right). \tag{16}$$

4. Anti-flatness and combined objective

The anti-flat gate is

$$s_{\mathrm{nonflat}} = \min\left(1,\ \frac{\max|\chi - 1| + \max|\beta^x|}{\tau}\right), \tag{17}$$

with scale $\tau \sim 0.08$, giving

$$F_{\mathrm{FTL}} = \max(F_{\mathrm{null}}, F_{\mathrm{portal}}, F_{\mathrm{throat}}) \cdot s_{\mathrm{nonflat}}. \tag{18}$$

The optimizer uses $F_{\log}$ rather than raw F_{FTL}.

5. Validation: FTL scoring across reference seeds

Table II checks the reference seeds at $L = 8$.

The log term makes subluminal Alcubierre signals visible, while throat-like seeds score through proper-distance and asymmetry channels even when $F_{\mathrm{null}} = 0$.

TABLE I. Compact score-weight summary. Components are inactive when their diagnostics are unavailable.

Group	Weights
FTL value	$5F_{\text{log}} + 2F_{\text{asymmetry}} + s_{\text{nonflat}} + 3F_{\text{op}}$
Health	$1.5s_{\text{survival}} + 2s_{\text{constraints}} + s_{\text{lapse}} + 2.5s_{\text{comoving}} + 0.5s_{\text{stability}}$
Risk penalties	$1.5s_{\text{horizon}} + 2s_{\text{EC}} + 2s_{\text{growth}} + 1.5s_{\text{anec}} + s_{\text{tidal}}$
Evolved geometry	$0.5F_{\text{curv}} + 1.5s^{\text{eff}}_{\text{exotic}}$

TABLE II. FTL metric scores for reference spacetime seeds ($L = 8$). Raw $F_{\text{FTL}} = \max(F_{\text{null}}, F_{\text{portal}}, F_{\text{throat}}) \cdot s_{\text{nonflat}}$; F_{log} from Eq. 7 with $\eta = 100$.

Seed	F_{null}	F_{FTL}	F_{log}	F_{asym}	s_{nonflat}
Flat Minkowski	0	0	0	0	0
Alcubierre $v = 0.3$	0.063	0.063	0.432	0.083	1.0
Alcubierre $v = 0.5$	0.094	0.094	0.508	0.083	1.0
Alcubierre $v = 0.8$	0.131	0.131	0.573	0.083	1.0
Ellis–Bronnikov (16 bases)	0	0.945	0.988	0.998	1.0
Schwarzschild	0	0.878	0.972	0.999	1.0

TABLE III. CMA-ES search dimensions with bounds. All coefficients are dimensionless amplitudes for the Gaussian basis expansion.

Parameter	Lower	Upper	Initial
`recipe_chi_coeff_0`	−0.5	0.1	−0.1
`recipe_chi_coeff_1`	−0.3	0.3	0
`recipe_chi_coeff_2`	−0.2	0.2	0
`recipe_chi_coeff_3`	−0.2	0.2	0
`recipe_basis_width`	0.3	3.0	1.0
`recipe_alpha_coeff_0`	−0.3	0.3	0
`recipe_alpha_coeff_1`	−0.3	0.3	0
`recipe_beta_coeff_0`	−0.5	0.5	0
`recipe_beta_coeff_1`	−0.5	0.5	0

6. *Why FTL scoring resolves the flat-space attractor*

The three shortcut channels are independent: shift-driven warp, $\chi > 1$ compression, and areal-radius throat pinching. The nine-dimensional default search vector in Table III spans these mechanisms while remaining tractable.

7. *GPU validation with upgraded FTL scoring*

Completed GPU smoke episodes were re-scored with the upgraded objective.

a. Reference seeds: flat-space attractor resolved. Table IV shows the key result: Alcubierre $v = 0.5$ outranks flat Minkowski because F_{log} and s_{comoving} offset the Eulerian stability penalty.

The same scoring keeps random χ-only angular candidates below validated warp/throat seeds.

b. Non-spherical angular candidates. Table V shows that non-spherical χ deformations produce strong asymmetry but no axial null shortcut.

These campaigns establish the baseline: shift-enabled warps can exceed flat space, while angular χ deformations plateau on asymmetry alone.

E. Dynamical stability scoring (implemented)

Short runs can survive while already focusing toward collapse. We therefore score geometry growth explicitly instead of relying on survival time alone.

1. *Data sources*

The metric uses `collapse_diagnostics.dat` and `areal_radius.dat`: $\alpha_{\min}$, $\chi_{\min}$, $|K|_{\max}$, r_H, and R_{areal} over the full run.

2. *Fractional violation terms*

For each sampled diagnostic,

$$\Delta q_{\text{grow}} \equiv \frac{\max_t q(t) - q(0)}{\max\big(|q(0)|,\, q_{\text{floor}}\big)},$$
$$\Delta q_{\text{drop}} \equiv \frac{q(0) - \min_t q(t)}{\max\big(|q(0)|,\, q_{\text{floor}}\big)}, \tag{19}$$

with terms clipped to ≥ 0. The implemented indicators are:

$$\Delta_K \equiv \Delta\big(|K|_{\max}\big)_{\text{grow}} \quad \text{with } q_{\text{floor}} = 10^{-1}, \tag{20}$$
$$\Delta_\alpha \equiv \Delta\big(\alpha_{\min}\big)_{\text{drop}} \quad \text{with } q_{\text{floor}} = 10^{-6}, \tag{21}$$
$$\Delta_\chi \equiv \Delta\big(\chi_{\min}\big)_{\text{drop}} \quad \text{with } q_{\text{floor}} = 10^{-8}, \tag{22}$$
$$\Delta_{r_H} \equiv \Delta\big(r_H\big)_{\text{grow}} \quad \text{with } q_{\text{floor}} = 1, \tag{23}$$
$$\Delta_{R_{\text{areal}}} \equiv \max_t \frac{\big|R_{\text{areal}}(t) - R_{\text{areal}}(0)\big|}{\max\big(|R_{\text{areal}}(0)|,\, 10^{-8}\big)}. \tag{24}$$

Missing diagnostics are omitted.

TABLE IV. GPU smoke results with upgraded FTL scoring ($t = 2$, $N_{\text{full}} = 64$, $L = 8$). Scores recomputed from archived episodes with `score.py` weights from Table I.

Candidate	$F_{\log}$	F_{asym}	s_{comoving}	$s_{\text{stability}}$	S
Alcubierre $v = 0.5$	0.508	0.083	0.998	0.02	**10.08**
Flat Minkowski	0	0	1.000	1.00	10.00
Ellis–Bronnikov (N=16)	0.988	0.998	0.074	0.07	**10.84**
`random_angular_004`	0	0.989	0.019	0.50	6.43
`random_000`	0	0.391	0.015	0.46	3.91
`random_003`	0.174	0.476	0.023	0.02	3.76

TABLE V. Non-spherical GPU batch with upgraded FTL scoring ($t = 2$, $N_{\text{full}} = 64$, stamp `20260528_151713`).

Candidate	$F_{\log}$	F_{asym}	s_{comoving}	$s_{\text{stability}}$	S
`random_angular_004`	0	0.989	0.019	0.50	6.43
`random_angular_007`	0	0.989	0.020	0.51	6.13
`quadrupole_bubble_001`	0	0.989	0.019	0.50	6.14
`random_angular_005`	0	0.989	0.019	0.49	6.14
`random_angular_006`	0	0.989	0.019	0.49	6.15
`dipole_lopsided_000`	0	0.989	0.015	0.44	6.16
`mixed_modes_002`	0	0.989	0.015	0.44	6.12

3. Geometry-change budget and stability reward

The budget is

$$\Delta_{\text{geom}} = \Delta_K + \Delta_\alpha + \Delta_\chi + \Delta_{r_H} + \Delta_{R_{\text{areal}}}. \tag{25}$$

and the reward is

$$s_{\text{stability}} = r(\Delta_{\text{geom}}; 1) = \frac{1}{1 + \Delta_{\text{geom}}}. \tag{26}$$

Higher is better; $s_{\text{stability}} < 0.25$ is treated as a promotion failure.

4. Promotion criteria

Promotion requires survival, controlled constraints, and $s_{\text{stability}} > 0.25$; a good equilibrium should approach $s_{\text{stability}} \gtrsim 0.8$.

F. Longer-run validation: exotic matter, evolved-frame FTL, and a visualization caveat

Longer $t = 16$ runs test the exotic scalar matter fix and evolved operational FTL probe beyond the smoke window.

The results have three main implications. First, the exotic field now produces the expected matter NEC violations for Ellis–Bronnikov and mild non-spherical phantom candidates. Second, effective energy-condition evaluation of $T^{\text{eff}}_{\mu\nu} = G_{\mu\nu}/8\pi$ recovers the geometric exotic energy of shift-driven Alcubierre data even when scalar-matter NEC is zero. Third, evolved-frame FTL is essential: Alcubierre's initial shortcut decays, while Ellis–Bronnikov develops an evolved shortcut absent on the $t = 0$ slice. The gated objective removes flat Minkowski without changing structured candidates.

VIII. CMA-ES VS. REINFORCEMENT LEARNING

The current pipeline is CMA-ES, not reinforcement learning: it samples continuous coefficients, ranks them by $-S(\theta)$, and updates a Gaussian search distribution. RL becomes relevant only for sequential decisions such as choosing topology, modes, or mid-run gauge actions.

A. Hierarchical search: topology vs. geometry

CMA-ES cannot add mouths, throats, or new mode families. A future two-level search would let a discrete controller choose topology and active modes, then hand the resulting skeleton to CMA-ES/MAP-Elites for continuous refinement, echoing discrete/continuous decompositions in string-landscape search [3].

TABLE VI. Longer-run validation ($t = 16$, $N_{\rm full} = 64$, $L = 8$, H100) with the gated objective (Eq. 11). $F^0_{\rm op}/F^{\rm ev}_{\rm op}$ are the operational FTL benefits on the $t = 0$ slice and the evolved plotfile; $\mathrm{NEC}^{\rm mat}$ is the minimum matter null-energy margin over the run (negative = violation); $g_{\rm nt}$ is the non-triviality gate. S is the gated total: flat Minkowski collapses to 0 while every non-trivial geometry is unchanged. All reference runs use constrained phantom initial data.

Example	survival	$s_{\rm stab}$	$F^0_{\rm op}$	$F^{\rm ev}_{\rm op}$	$\mathrm{NEC}^{\rm mat}$	$g_{\rm nt}$	S
Flat Minkowski	1.00	1.00	0	—	0	0.0	**0.0**
Alcubierre $v = 0.5$	1.00	0.02	0.096	0.0005	0	1.0	**14.9**
Ellis–Bronnikov (N=16)	0.80	0.08	0	0.042	−0.62	1.0	12.8
`quadrupole_bubble_001`	1.00	0.18	0	0.0095	−0.013	1.0	10.5
Schwarzschild puncture	0.29	0.00	0	—	−0.91	1.0	9.7

IX. AVOIDING FAKE FTL (OPERATIONAL PROBE TESTS)

A major risk is gauge-driven fake FTL. High-scoring candidates must therefore pass an operational arrival-time test against a flat control on the same grid, with no trapped-surface crossing and bounded constraints.

X. IMPLEMENTED: CONSTRAINT-SATISFYING METRIC GUESSER

For the conformally flat recipe ($\tilde\gamma_{ij} = \delta_{ij}$, $A_{ij} = 0$), the constraints reduce to

$$R[\chi] + \tfrac{2}{3}K^2 = 16\pi\rho, \tag{27}$$

$$-\tfrac{2}{3}D^i K = 8\pi j^i. \tag{28}$$

With $\Pi = 0$, the momentum constraint requires spatially constant K; ϕ is then derived from χ to reduce Hamiltonian violation.

A. Pipeline components

The implemented components are: `constrained_recipe.py` (derive ϕ from χ with K = const, $\Pi = 0$), `preflight.py` (reject large residuals or negative χ before GPU launch), `RadialRecipeLevel.cpp` (write $\rho_{\rm req}$ and constraint diagnostics), `score.py`/`optimize.py` (CMA-ES scoring), `seeds.py` (Minkowski, Ellis–Bronnikov, Schwarzschild), plotfile extraction, and batch GPU scripts.

B. Verification and validation

A test suite passes **74/74 assertions**: flat space remains exact, $K = \text{const}, \Pi = 0$ drives $\|M_i\|_{L^2} = 0$, pathological inputs are rejected, and constrained overrides usually reduce $\|H\|_{L^2}$. On 27 synthetic profiles, the batch harness finds 7 `accepted_phantom`, 19 `rejected_exotic_energy`, and 1 `rejected_negative_chi`.

C. First non-spherical (symmetry-breaking) step

The first symmetry-breaking extension adds axisymmetric deformations,

$$\chi(r,\theta) = \chi_0(r) + \sum_a A_a\, e^{-(r-r_a)^2/2w_a^2}\, P_{\ell_a}(\cos\theta), \tag{29}$$

with a ray check to ensure $\chi > 0$. The seed-42 run produced 7 ray-sane candidates and one negative-χ rejection.

D. GPU/C++ RadialRecipe smoke validation

The Python wrapper and `Examples/RadialRecipe/` executable now run end-to-end on H100 GPUs: generate `params.txt`, check parameters, evolve, and parse diagnostics.

1. *Radial smoke results at t=2*

Table VII summarizes the first GPU handoff campaign; labels indicate short-run survival only.

The 3D diagnostics stay controlled even when the coarse 1D pre-flight residual is conservative. Extended $t = 5$ tests promote Ellis–Bronnikov and `bubble_wall_016` to the resolution ladder.

2. *Resolution note (Ellis–Bronnikov)*

Increasing $N_{\rm full}$ from 64 to 128 sharpens the Ellis–Bronnikov throat while keeping constraints controlled.

3. *AMR closed-loop test*

The first full-domain AMR CMA-ES run found only low-stability survivors (Table VIII), confirming the stop-time exploit that the growth metric targets.

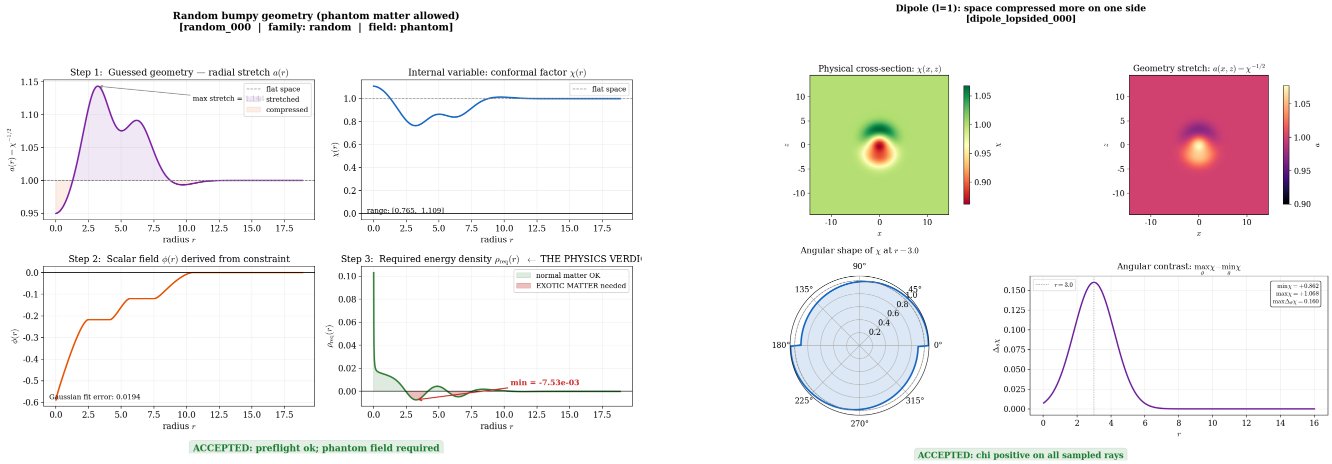

FIG. 1. **Metric-guesser validation examples.** Left: accepted phantom random candidate, showing guessed stretch $a(r) = \chi^{-1/2}$, stored $\chi(r)$, derived scalar $\phi(r)$, and required energy density $\rho_{\rm req}(r)$; it is numerically constructible but accepted only in the phantom/exotic-matter class because $\rho_{\rm req} < 0$ in places. Right: accepted non-spherical dipole candidate; the $\ell = 1$ mode makes the geometry lopsided while χ stays positive throughout the ray check.

TABLE VII. Short GPU smoke results for spherically symmetric RadialRecipe candidates ($t = 2$, $N_{\rm full} = 64$, `max_level=0`). $s_{\rm stability}$ and S are from the eight-component scorer (May 2026 rescoring). "Short-run survivor" means controlled diagnostics over this window, not long-time stability.

ID	Python $\|H\|_{L^2}$	C++ max $\|H\|_{L^2}$	$\alpha_{\min}$	$\chi_{\min}$	$s_{\rm stability}$	S	Outcome
`ellis_bronnikov`	2.24	0.010	0.66	0.83	0.30	4.33	Short-run survivor
`bubble_wall_016`	0.027	0.004	0.96	1.00	0.68	6.35	Short-run survivor
`random_000`	0.224	0.042	0.54	0.77	0.46	3.92	Short-run borderline

E. Non-spherical 3D GPU batch

All seven ray-sane non-spherical candidates survive short GPU evolution with controlled constraints (Table IX).

Their stability scores exceed the AMR-optimized radial survivors but remain below true-equilibrium values.

F. Gauge-sector angular search and streaming HQ validation (implemented)

Because angular χ cannot be made constraint-consistent by a 1D solve, we next opened the gauge sector: α and β^i carry Legendre modes with $\ell \in \{1, 2\}$ and searchable radial centers/widths, giving a 21-parameter gauge-angular space.

a. Search outcome. An 8-GPU campaign found evolved superluminal channels at $\int \rho_- \, dV \approx 0.13$–$0.17$, a factor 2–5 below spherical persistent-FTL candidates. The top candidate, `eval_000188`, was promoted.

b. Streaming high-quality validation. `eval_000188` was re-evolved to $t = 16$ at $N_{\rm full} = 192$, `max_level=2`, with on-the-fly plotfile consumption and deletion, keeping the full validation near 1.2 GB.

The validated geometry is a standing Alcubierre-like χ dipole, not a translating bubble: the evolved channel persists weakly by $t = 16$, constraints decrease, and exotic matter remains small and axial, but a near-trapped flag appears in the compressed lobe.

c. Why not a wormhole or portal. The current parameterization cannot change topology, and its non-spherical freedom lives in the gauge sector, where shift structure naturally favors warp-like channels. Wormholes and portals require topology search plus a 3D constraint solver.

XI. FORWARD PATH: FROM STABILITY GAP TO ANALYTIC DISCOVERY

The main remaining challenge is the dynamical stability gap: optimizers can still favor geometries that merely outlive a short stop time. The production path is:

A. Phase 1: Closing the stability gap (immediate software priority)

Use $s_{\rm growth}$ to reject positive growth, calibrate it against $t = 5$–10 ladders, keep the FTL scaffolding/gated objective active, and sweep basis size/width before pro-

TABLE VIII. Top AMR CMA-ES candidates (`optimize_20260528T055206Z`, $t = 2$, stability-aware scoring).

Eval	S	$s_{\text{stability}}$	$\alpha_{\min}$	$\chi_{\min}$	$\lvert K\rvert_{\max}$
`eval_000013`	4.12	0.085	0.29	0.47	1.02
`eval_000006`	4.05	0.056	0.24	0.41	2.33
`eval_000009`	4.00	0.054	0.19	0.35	2.39
`eval_000002`	3.87	0.049	0.07	0.29	1.84
`eval_000015`	2.21	0.035	0.00	0.06	1.73

TABLE IX. Non-spherical GPU smoke results ($t = 2$, $N_{\text{full}} = 64$, stability-aware scoring).

Candidate	$\max \lVert H\rVert_{L^2}$	$\alpha_{\min}$	$\chi_{\min}$	$s_{\text{stability}}$	S
`dipole_lopsided_000`	0.0118	0.90	0.88	0.44	4.93
`quadrupole_bubble_001`	0.0125	0.92	0.87	0.50	4.98
`mixed_modes_002`	0.0118	0.91	0.87	0.44	4.89
`random_angular_004`	0.0064	0.92	0.87	0.50	5.31
`random_angular_005`	0.0122	0.91	0.87	0.49	4.98
`random_angular_006`	0.0132	0.91	0.87	0.49	4.97
`random_angular_007`	0.0135	0.92	0.87	0.51	4.98

TABLE X. High-quality streaming validation of the gauge-angular winner `eval_000188` ($N_{\text{full}} = 192$, `max_level=2`, $t = 16$, H100). Initial vs. evolved operational FTL, exotic matter, and constraint health.

Quantity	Value
$\max c$ (initial slice)	2.67
$\max c$ (evolved, $t = 16$)	1.32
F^{0}_{op} / $F^{\text{ev}}_{\text{op}}$	0.35 / 5.6×10^{-3}
superluminal path fraction (evolved)	0.51
$\int \rho_{-}\, dV$ (exotic matter)	0.134
ANEC line integral (axis)	-6.6×10^{-3}
$\lVert H\rVert_{L^2}$ (max / final)	1.1×10^{-3} / 4.2×10^{-4}
$\lVert M\rVert_{L^2}$ (final)	3.6×10^{-5}
$\min \theta_{+}$ (near-trapped flag)	-0.11
S (gated total)	16.75

motion.

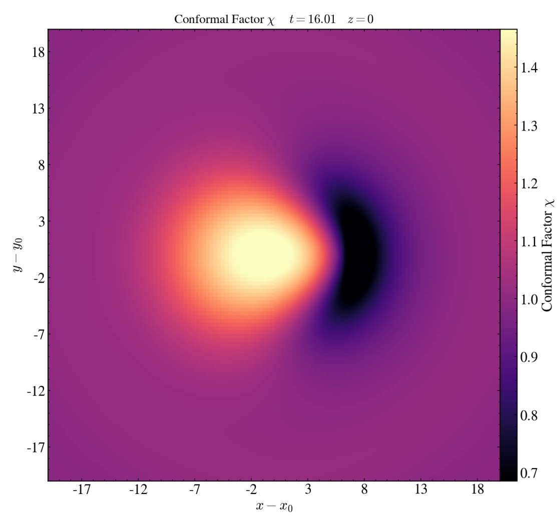

FIG. 2. **Persistent directional channel of the gauge-angular winner `eval_000188`** (conformal factor χ, $z = 0$ slice, $t = 16$, $N_{\text{full}} = 192$). A standing front–back dipole compresses space on the $+x$ side ($\chi \approx 0.7$, dark) and expands it behind ($\chi \approx 1.45$, bright)—an Alcubierre-like contract-ahead/expand-behind polarity realized through the shift sector at low exotic-matter cost. The structure does not translate; it stands and slowly decays. Frames were extracted on the fly during evolution while plotfiles were deleted (keep-last-2).

B. Directional non-spherical search: next degrees of freedom

Gauge-angular search produced a low-exotic directional channel. Next degrees of freedom are:

1. searchable radial placement of angular modes;
2. full-z domains without reflective parity;
3. persistence/co-motion rewards;
4. soft barriers on near-trapped surfaces;
5. higher angular order and a dedicated translation shift.

a. The geometric sector. The largest improvement is a 3D elliptic constraint solver, allowing directional structure in χ or $\tilde{A}_{ij}$ without reintroducing off-axis constraint error.

b. Topology search. Wormholes and portals require a discrete controller that selects mouths, throats, modes, and boundary connectivity before the continuous optimizer refines each topology.

C. Phase 2: Multi-observer energy conditions

The $t = 0$ ANEC and tidal proxies are blind to pure-shift warps, so `warpfactory.py` evaluates energy conditions from the evolved 4-metric via $T_{\mu\nu} = G_{\mu\nu}/8\pi$, e.g.

$$\Xi_W(\mathbf{x}) = \min_V T_{\mu\nu} V^\mu V^\nu, \qquad g_{\mu\nu} V^\mu V^\nu = -1, \tag{30}$$

This recovers Alcubierre NEC/WEC violations missed by χ-sourced proxies.

The evaluator also uses a Hawking–Ellis/Type-I hybrid margin following WARPAX [6], with fourth-order finite differences [7]; its margin feeds `score.py` as $s^{\text{eff}}_{\text{exotic}}$.

D. Phase 3: GPU-cluster production search

With surrogate screening, MAP-Elites, and Pareto extraction in place, production runs are a 1000-candidate constraint atlas plus a 50-generation CMA-ES campaign:

```
python -m grteclyn_wrapper \
    --example RadialRecipe \
    --constrained --preflight optimize \
    --max-generations 50 --seed 42
```

E. Phase 4: Analytical back-derivation

A numerical optimum is only a coefficient list. The final handoff reconstructs χ_{num}, fits a closed form by symbolic regression, solves the Hamiltonian constraint for $\phi(r)$, and reloads the analytic profile for verification. This can be automated as a multi-agent loop inspired by PHYSAGENT [4].

XII. EXTENSIONS: IMPLEMENTED METRICS, SEARCH, AND ROBUSTNESS

The extensions below target three limits: short-run gaming, coordinate-dependent energy/FTL verdicts, and expensive GPU evaluations.

A. New physical metrics

a. Constraint growth rate. Fit late-time diagnostics to $q(t) \sim q_0 e^{\lambda t}$ and penalize the worst growth:

$$\begin{aligned} s_{\text{growth}} &= r(\max(0, \lambda_{\text{eff}});\, \sigma_\lambda)\,, \\ \lambda_{\text{eff}} &= \max\!\big(\lambda_{\|H\|},\, \lambda_{|K|_{\max}},\, \lambda_{1/\chi_{\min}}\big), \end{aligned} \tag{31}$$

with $\sigma_\lambda = 0.5$.

b. ANEC line proxy. Integrate required energy along the travel axis:

$$\begin{aligned} \mathcal{A}_{\text{line}} &= \int_{-L}^{L} \rho_{\text{req}}(x)\, dx, \\ s_{\text{anec}} &= r(\max(0, -\mathcal{A}_{\text{line}});\, \sigma_{\mathcal{A}})\,, \end{aligned} \tag{32}$$

with $\rho_{\text{req}} = (R + \frac{2}{3}K^2)/16\pi$ and $\sigma_{\mathcal{A}} = 0.1$. The full target is

$$\mathcal{A} = \int_\gamma T_{\mu\nu}\, k^\mu k^\nu\, d\lambda, \tag{33}$$

which requires the evolved probe and a local tetrad.

c. Tidal proxy. Use $T_{\text{tidal}} = \max_x |R(x)| + \max_x |\partial_x^2 \alpha(x)|$ and a $t = 0$ trapped-surface flag until full 4-curvature diagnostics are available.

d. Operational probe FTL. Compare first arrival on the candidate and an identical flat control:

$$F_{\text{op}} = \frac{t_B^{\text{flat}} - t_B^{\text{curved}}}{t_B^{\text{flat}}}, \tag{34}$$

accepted only with no trapped-surface crossing and bounded constraints.

Table XI gives reference $t = 0$ proxy values.

Table XII shows that every short-run survivor still has positive λ_{eff}, exposing the stability gap from a single smoke run.

B. Optimizing the search

a. Surrogate-assisted screening. `surrogate.py` uses an RBF kernel-ridge model to pre-screen CMA-ES proposals; one 8-GPU run skipped 27% of GPU evaluations while retaining the best score.

b. Quality-Diversity. `qd_search.py` maintains MAP-Elites over FTL benefit and exoticity, producing a diverse failure atlas rather than a single optimum.

c. Pareto front. `pareto.py` reports non-dominated candidates over $(F_{\text{FTL}}, s_{\text{anec}}, s_{\text{growth}}, s_{\text{tidal}})$, exposing trade-offs hidden by scalar weights.

d. Constraint-solved initial data. Making elliptic Path B the default is the largest single improvement: every proposal would start constraint-satisfying, reducing junk radiation and making growth rates cleaner.

TABLE XI. Implemented $t = 0$ gauge-robust proxies on reference seeds ($L = 8$, 1024 points). $\mathcal{A}_{\text{line}}$ is the axis energy-density line integral (Eq. 32); $\mathcal{K}^{1/2}_{\text{proxy}} \equiv \max|R|$ is the spatial-curvature scale; "trap" is the $t = 0$ trapped-surface flag.

Seed	$\mathcal{A}_{\text{line}}$	s_{anec}	$\max\|R\|$	s_{tidal}	trap
Flat Minkowski	0	1.000	0	1.000	no
Alcubierre v=0.5	0	1.000	0	1.000	no
Ellis–Bronnikov	−6.62	0.015	$2.0{\times}10^4$	0.000	yes
Schwarzschild	−2.38	0.040	$6.0{\times}10^3$	0.000	yes

TABLE XII. GPU growth-rate and new-component values ($t = 2$, $N_{\text{full}} = 64$, `max_level=0`). λ_{eff} from Eq. (31); S is the total under the extended 15-component scorer (adds $s_{\text{growth}}, s_{\text{anec}}, s_{\text{tidal}}$ to Table I).

Candidate	λ_{eff}	s_{growth}	s_{anec}	s_{tidal}	S
`alcubierre_warp`	0.99	0.335	1.000	1.000	13.25
`ellis_bronnikov`	1.02	0.329	0.015	0.000	12.18
`bubble_wall_016`	0.41	0.550	0.987	0.524	10.76
`random_000`	0.29	0.633	0.682	0.161	7.57

TABLE XIII. MAP-Elites archive from one 40-evaluation, 8-GPU campaign (6×6 grid, coverage 0.19). Each row is the elite of a distinct behavior cell.

Cell	FTL benefit	Exoticity	Elite S
$(1, 0)$	0.319	0.000	12.47
$(3, 1)$	0.618	0.248	12.13
$(2, 0)$	0.384	0.000	11.68
$(3, 0)$	0.595	0.040	11.62
$(0, 0)$	0.000	0.000	10.98
$(1, 1)$	0.332	0.228	10.21
$(0, 1)$	0.160	0.231	9.91

XIII. CONCLUSION

This work turns exotic-metric study into a closed-loop GPU search over dynamical initial data. The current pipeline proposes constrained recipes, evolves them in `GRTeclyn`, scores stability/FTL/energy diagnostics, and validates promising candidates on longer runs. The first campaigns expose the central stability gap but also show that FTL scaffolding, non-triviality gating, growth metrics, evolved energy-condition checks, surrogate screening, MAP-Elites, and Pareto extraction can steer the search away from trivial flat space. The next step is a 3D elliptic constraint solver and topology-aware search, enabling genuinely geometric warp, portal, and wormhole candidates to be tested with observer-robust physical diagnostics.

[1] S. Alexander, B. Bradley, L. Gouskos, and C. Niu, "Autonomous Discovery of Particle Physics Theories from Experimental Data," arXiv:2603.28935 [hep-ph] (2026).

[2] S. Qiu, Z. Cai, J. Wei, Z. Li, Y. Yin, Q.-H. Cao, C. Liu, M.-x. Luo, X.-B. Yuan, and H. X. Zhu, "An End-to-end Architecture for Collider Physics and Beyond," arXiv:2603.14553 [hep-ph] (2026).

[3] J. Halverson, B. Nelson, and F. Ruehle, "Branes with Brains: Exploring String Vacua with Deep Reinforcement Learning," JHEP **06**, 003 (2019), arXiv:1903.11616 [hep-th].

[4] X.-Q. Han, Z.-F. Gao, P.-J. Guo, and Z.-Y. Lu, "PhysAgent: A Multi-Agent Approach to Automated Discovery of Physical Laws," arXiv preprint (2025).

[5] C. Helmerich, J. Fuchs, A. Bobrick, L. Sellers, B. Melcher, and G. Martire, "Analyzing Warp Drive Spacetimes with Warp Factory," Class. Quantum Grav. (2024), arXiv:2404.03095 [gr-qc].

[6] A. T. Le, "Observer-robust energy condition verification for warp drive spacetimes," arXiv:2602.18023 [gr-qc] (2026).

[7] C. O. Lousto, "A time-domain fourth-order-convergent numerical algorithm to integrate black hole perturbations in the extreme-mass-ratio limit," Class. Quantum Grav. **22**, S543 (2005), arXiv:gr-qc/0503001.

[8] Z.-H. Li, C.-Q. Li, and L.-G. Pang, "Solving Einstein equations using deep learning," arXiv:2309.07397 [gr-qc] (2023).